Learning to Allocate Risk in Options Markets

Contextual Position Sizing and Q-Learning for Capital Allocation

Adam Kreitzman and Soo Xin Yang

20 October 2026

Abstract

We study two exposure decisions in equity options: contextual position sizing within strategies and capital allocation between strategies with fitted Q-learning. Contract-level simulations separately construct portfolios of approximately 30-day, 25-delta put pairs and short ATM straddles selected by persistent realized-minus-implied volatility. A separate allocator reallocates capital between fractional claims on the funded strategy return indices, rather than a shared contract inventory. During January 2022–August 2025, learned entry sizing raises annualized excess Sharpe from 1.055 to 1.263 for put pairs and from 0.990 to 1.041 for straddles, relative to full sizing. At the illustrative risk preference $\lambda = 0.20$, combining both learners raises Sharpe from 1.088 to 1.726 in a four-way comparison. The original validation-selected Q policy instead achieves later Sharpe 0.282, so the illustrative gain does not establish successful validation-based policy selection. With learned-size entries, Q also exceeds rolling mean–variance allocation: 1.726 versus 1.519. Interpretation connects entry decisions to completed profitability and volatility, and strategy allocation to the changing reward for straddle risk. Statistical uncertainty and assumed execution quality qualify these historical results.

1 Introduction

Finding a useful trading signal leaves an investor with a second problem: how much capital should it receive as opportunities and risks change? In options markets, a strategy with favorable risk-adjusted returns can deploy little capital, while a strategy that earns more dollars can expose the account to much larger movements. A ranking of average returns or Sharpe ratios alone does not determine when either strategy deserves funding. We study whether information available at the time of a decision can improve that allocation.

We make the question concrete with two equity-options strategies. The first buys relatively cheap puts and sells relatively expensive puts on related names, using approximately 30-day, 25-delta contracts. It matches vega at entry and adjusts stock delta hedges daily, while retaining differences in realized paths, jumps and company-specific events. The second sells approximately 30-day ATM straddles on names whose completed historical outcomes show persistently low realized volatility relative to entry implied volatility. It also hedges delta daily, but retains net short-gamma and short-vega exposure. These are different sources of option risk: relative pricing across names and premium income against future movement.

Our central learning question is how to allocate capital between the resulting portfolios. A fitted Q-learning policy chooses weekly capital targets using current opportunity measures, completed strategy performance, risk conditions and previous holdings. Entry sizing supplies a supporting decision within each strategy: contextual sizing policies choose 50%, 75% or 100% of a proposed trade's full size. These sizing policies are fitted in batches to completed outcomes and optimize aggregate profit paths; they do not implement an online exploration-and-reward update. The economic rules determine proposals, directions and contracts; the entry policies alter commitments

and subsequent admission capacity, while the allocation policy changes the capital assigned to each funded strategy index.

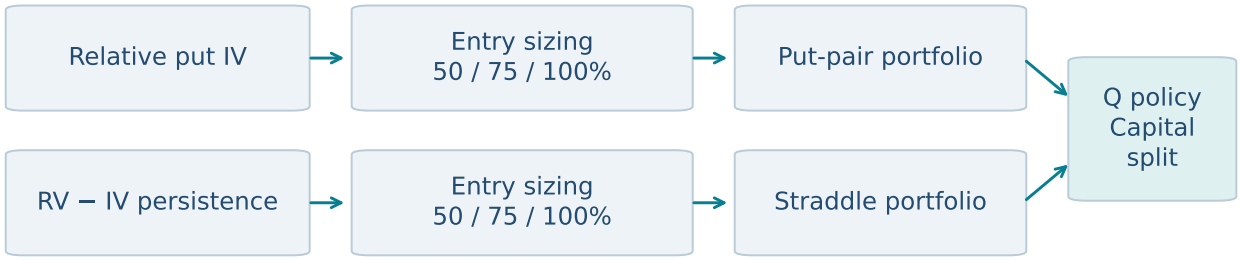


Figure 1. Two exposure decisions. Entry policies size proposals within each signal; a separately fitted Q policy chooses the capital split between funded strategy indices.

We evaluate these decisions through actual-contract simulations with option marks, daily stock hedges, financing and explicit trading charges. A four-way comparison evaluates full-size entries with fixed 50/50 allocation, learned-size entries with fixed 50/50 allocation, full-size entries with Q allocation, and learned-size entries with Q allocation. All four receive the same signal proposals, and Q is independently refitted for each entry-policy family. During January 2022–August 2025, annualized excess Sharpe rises from 1.088 with full-size entries and fixed 50/50 allocation to 1.726 with both learners at the illustrative risk preference $\lambda = 0.20$. The original validation-selected policy has later Sharpe 0.282; the headline comparison illustrates allocation behavior rather than successful transfer of the validation selection. With learned-size entries, Q also exceeds rolling mean–variance allocation, 1.726 versus 1.519. Timing and one-period controls investigate whether the allocation result reflects average exposure, conditional information or continuation value; they indicate a larger role for conditional allocation and a smaller incremental sequential benefit. These comparisons describe a profit–risk tradeoff, with no single policy dominating every entry size, risk preference or performance measure.

Both 2018–2021 validation and January 2022–August 2025 evaluation remain visible throughout. Individual fits use strictly completed past outcomes; whether and how later-period results informed development remains under verification in Section B.1. The evaluation is treated as researched historical evidence rather than a confirmed independently frozen test of the complete research process. Statistical uncertainty and assumed execution quality qualify the performance gains. The outer allocator also trades fractional claims on separately funded strategy indices; it does not rebuild a shared whole-contract inventory. Our contribution is a controlled examination of adaptive exposure allocation under these assumptions, motivating integrated implementation and prospective evaluation.

2 Related Work

Ning and Lee [1] combine equity-spread construction with tabular Q-learning for entry and exit. They select hedge coefficients by minimizing empirical mean-reversion time and train the trading policy on simulated Ornstein–Uhlenbeck spreads. Evaluation on ten US equity pairs, formed in 2022 and traded in 2023, yields higher reported daily Sharpe ratios than their distance-method and Ornstein–Uhlenbeck benchmarks. Their analysis focuses on entry and exit within a constructed equity spread; it does not examine capital allocation across distinct options strategies. We study

that allocation decision using relative-value put pairs and short ATM straddles, with controlled comparisons that separate learned entry sizing from learned strategy allocation.

Bell et al. [2] develop AlphaGlass, an interpretable framework that maps stock characteristics into additive signals with sparse interactions and converts those signals into long–short portfolio weights through a differentiable rank-and-mask layer. The model is trained against portfolio objectives such as Sharpe or mean–variance utility, linking the learned signal directly to the investor's objective. Its additive structure makes portfolio weights traceable to individual characteristics and interactions. This combination of objective-based learning and interpretable decisions connects to our completed-path Sharpe optimization and policy diagnostics. The decision structures differ: AlphaGlass learns characteristic-based equity portfolio weights, whereas our fixed options signals generate candidate trades, contextual policies size those trades, and fitted Q-learning allocates capital across the resulting strategy portfolios.

Halperin [3] introduces QLBS, a discrete-time framework for option pricing and hedging based on a risk-adjusted Markov decision process. Q-learning jointly determines option values and hedge decisions by optimizing the risk-adjusted performance of a replicating portfolio, with a data-driven formulation that can learn from observed paths. QLBS provides a precedent for Q-learning in an options setting, but addresses a different control problem. Our option valuation and daily delta-hedging rules are prescribed; the learned sequential action is the capital allocation between two funded options-strategy indices.

3 Data and Backtesting Framework

3.1 Data and Sample

Historical option quotes, stock prices, interest rates and dividends come through WRDS. The prepared universe contains 125 pairs involving 161 equity/ETF underlyings; 46 pairs supply 608 screened put proposals over 2018–2025. The straddle proposal pool contains 35,774 candidates. The universe is retrospective and conditioned on usable coverage. Sections A.1 and A.4 record contract selection, quote filters and risk checks.

3.2 Execution and Account Construction

Each source strategy starts with $10 million. Signal-date contracts trade in the following session if quotes and account checks permit. At full size, put pairs target $450 vega per IV point on each leg; straddles target $450 across call and put together. Whole contracts are rounded; put pairs allow an initial vega mismatch of no more than 10%.

3.3 Valuation, Hedging, and Financing

Options normally remain until expiry. At each close, we mark the existing option and stock positions, record cash flows and then rebalance stock delta. The next day's hedge profit uses the stock held after that rebalance. Final option payoffs and hedge unwinds complete the trade. Trading the same name on later dates is possible subject to existing inventory and risk checks.

Stock transactions settle through cash, which earns the prior available benchmark rate; negative balances pay that rate plus one percentage point annually. Short-stock borrow costs 0.30% annually on short notional. Dividends enter as signed stock cash flows in addition to ex-dividend price movements.

Option fills use midpoint plus an adverse charge of 0.10 IV point times traded vega and $0.03 per contract; stock turnover costs one basis point. Expiry settlement has no option fill charge. Inventory, reserve, stress and quote checks determine admission. A request for less size can pass a pair limit and leave capacity for a later proposal. Section A.4 defines the checks; they are research assumptions rather than broker margin.

3.4 Evaluation Protocol

Results report 2018–2021 validation and January 2022–29 August 2025 evaluation. Each fit uses only outcomes completed before its fitting date; unfinished trade outcomes cannot enter training. The fitting windows and sizing procedures are specified with each strategy below. Accounts continue across the 2022 reporting boundary with accumulated cash and open positions.

Entry policies within each strategy receive the same signal proposals. Requested sizes nevertheless change contract rounding, outstanding risk and the capacity to admit later opportunities. The funded comparisons therefore measure complete sizing and admission policies. Standalone daily-profit paths used in entry-policy training differ from funded-account returns, which also reflect shared cash, inventory and admission constraints.

Source accounts begin with $10 million; the outer allocation experiment uses a separate single $10 million budget tracking the funded strategy indices. Appendix B records model fitting, with research-selection history in Section B.1. Appendix C collects supplementary results, including account-boundary diagnostics in Section C.4.

3.5 Performance Measures

Let N_t denote account NAV and K_t the continuing all-cash benchmark, both initialized at $10 million in January 2018. The benchmark compounds the previous available WRDS zero-curve rate nearest 30 days over calendar days using a 365-day year. For a reporting period, 0 is the preceding close and T the final close. Daily excess return and annualized excess Sharpe are

$$e_t = \frac{N_t}{N_{t-1}} - \frac{K_t}{K_{t-1}}, \qquad \widehat{S} = \sqrt{252}\,\frac{\overline{e}}{s_e},$$

where s_e is the sample standard deviation of daily excess returns. Annualized arithmetic mean excess return is $252\overline{e}$, and annualized excess volatility is $\sqrt{252}\,s_e$.

Dollar profit above cash measures the change in excess dollar wealth within the reporting period:

$$\Pi_{\text{excess}} = (N_T - K_T) - (N_0 - K_0).$$

With cash-relative wealth $W_t = N_t/K_t$ and Y equal to elapsed calendar days divided by 365.25, annualized cash-relative growth and maximum drawdown are

$$G_{\text{cash}} = \left(\frac{W_T}{W_0}\right)^{1/Y} - 1, \qquad D_{\text{cash}} = \max_{0\le t\le T}\left(1 - \frac{W_t}{\max_{0\le u\le t} W_u}\right).$$

The drawdown calculation includes the reporting period's opening close. Account return is $N_T/N_0 - 1$; its difference from the cash return is expressed in percentage points. Calendar-period account and cash returns are not annualized, including partial 2025. Annualized arithmetic mean excess return and compounded cash-relative growth are distinct measures. Accounts and the cash benchmark continue across reporting boundaries; profit curves subtract opening excess dollar wealth without liquidating carried positions.

4 Put Pair Strategy

The put-pair strategy trades relative implied volatility (IV) using approximately 30-day, 25-delta puts on related underlyings. Fixed rules determine trade eligibility, direction and contracts. A regression tree then assigns each proposal to a context, and cross-entropy method (CEM) search determines its entry size independently of the capital-allocation learner.

4.1 Trading Rules and Position Construction

Monthly formation restricts the candidate universe to pairs with 126-session return beta between 0.85 and 1.15 and correlation of at least 0.5. For each pair, we rank the IV difference $D_t = IV_{A,t} - IV_{B,t}$ against the preceding 252 trading sessions, requiring at least 63 observations. At the upper quintile we sell A's put and buy B's; at the lower quintile we reverse direction. These fixed percentile cutoffs measure relative richness against each pair's own spread history.

The proposed direction must also show positive historical profit per initial vega over both 91 and 365 calendar days, requiring at least 21 and 63 completed observations, respectively. These histories use completed simulated standalone candidates, including candidates not admitted to the funded account, and include trading and hedging costs; unfinished outcomes are excluded. The strategy therefore combines an unusual relative IV spread with directional profitability, rather than imposing unconditional spread mean reversion. These entry rules are shared across all put-sizing controls.

Full size targets $450 of vega per IV point on each leg, using integer option contracts. Vega is measured per one-percentage-point change in IV. Positions match vega at entry and rebalance stock delta daily; vega neutrality is not continuously restored, and gamma, jumps and company-specific risks remain. After entry, the simulator holds the purchased and sold contracts rather than replacing them when daily contract selection changes. Section A.1 records contract selection and Section A.4 records the admission constraints.

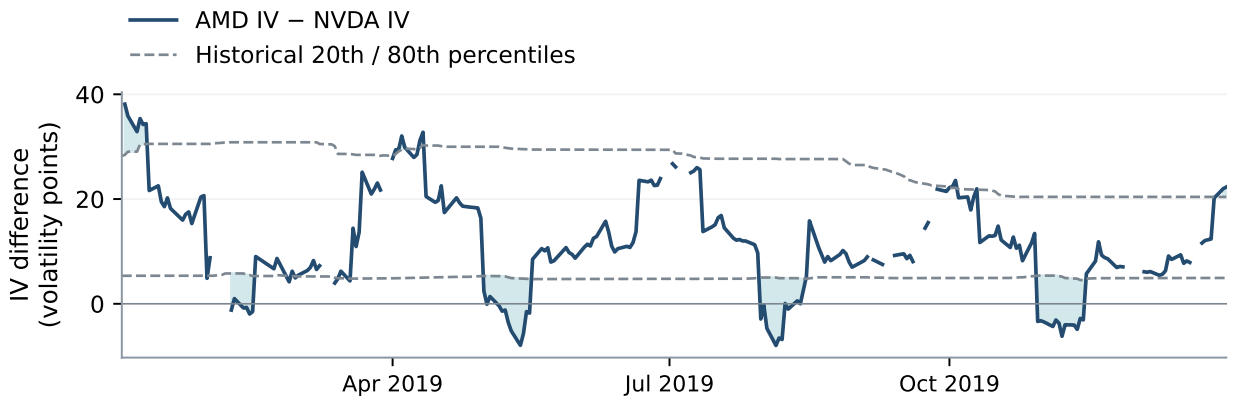


Figure 2. AMD minus NVDA selected-put IV in 2019. Shading marks excursions outside historical percentile boundaries, not executed trades. Gaps are unavailable quotes. Daily contract selection makes this a signal series, not the mark of one held contract.

The chart follows each day's selected contract rather than one held position. Profit depends on the executed contracts' option and hedge cash flows: a temporary narrowing of the displayed IV gap is neither required nor sufficient for a profitable expiry trade.

4.2 Decision-Tree Contexts

After the fixed rules determine a proposal, a regression tree maps its 27 inputs into at most eight leaf contexts. Each monthly fit uses completed examples from the preceding 18 months. Its target is cumulative full-size profit above cash, and splits reduce prediction error. With n examples, the minimum leaf count is $\max(8, \min(63, \lfloor n/8 \rfloor))$; fewer than 16 examples trigger full-size fallback. The tree defines contexts for sizing; it does not learn the trade-entry thresholds or direction.

The 27-input, 18-month specification was retained from 33 archived configurations ranked on 2018–2021 validation scores. Archived selection used a $345 million research account; the funded comparisons below use $10 million. Section B.1 records model selection, and Section B.2 gives additional estimation settings.

4.3 CEM Position Sizing

Given the fitted tree, CEM selects a map assigning 50%, 75% or 100% exposure to each leaf context. Both the tree and size map use completed put-pair outcomes, without the Q-allocation reward. Portfolio policies can be learned by optimizing investment objectives directly, including the Sharpe ratio [2]; here the objective is the Sharpe ratio of aggregate completed daily-profit paths.

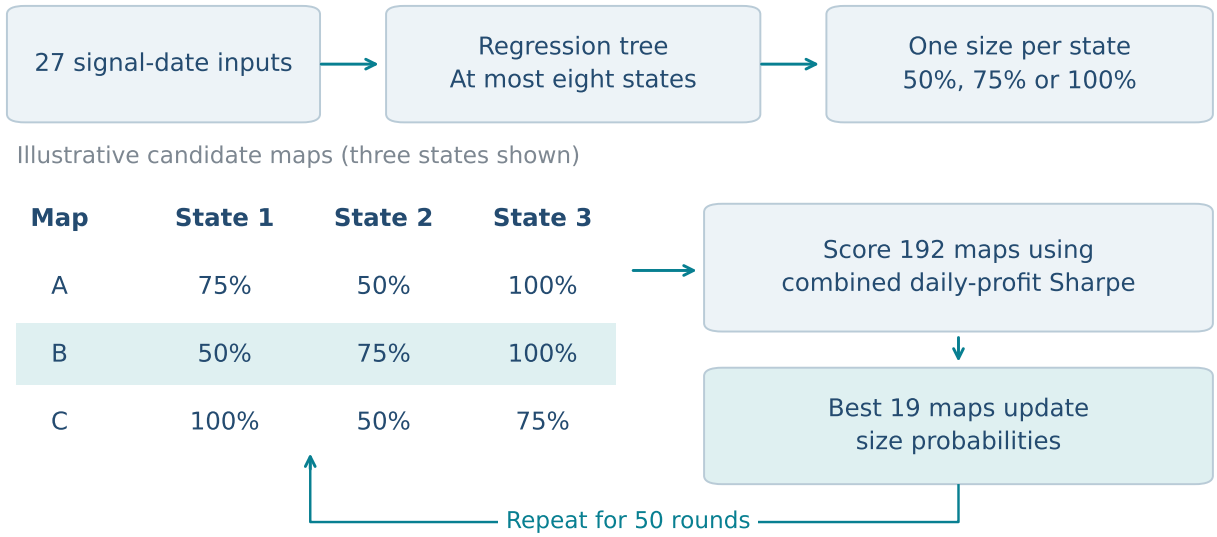


Figure 3. The original tree/CEM construction. The three-state maps illustrate candidates; actual trees have up to eight states. Each candidate selects action-specific completed paths; the best 19 update sampling probabilities. The retained map sizes next month's proposals.

For completed example i, $p_{i,d}(a)$ denotes profit above cash on day d at size a. Each size has its own rounded contracts and simulated hedge path. A map m assigns size $a_{\ell(x_i)}$ to the example's tree context. We sum its daily paths, retaining overlapping positions and zero-profit sessions, and score the result:

$$g_d(m) = \sum_i p_{i,d}(a_{\ell(x_i)}), \qquad \widehat{S}_{\text{train}}(m) = \sqrt{252}\,\overline{g}/s_g.$$

CEM [4] updates size probabilities toward the best 19 of 192 candidate maps for 50 rounds. Constant maps and the incumbent are anchors. The retained map sizes the following month's proposals.

This standalone-path training score differs from funded-account Sharpe: replay additionally applies shared cash, inventory and admission. Section B.2 gives exact fitting settings.

4.4 Results

4.4.1 Performance

Table 1. Put-pair entry-sizing policies in validation and later evaluation. S denotes annualized excess Sharpe; later profit is profit above cash.

Entry policy	2018–21 S	2022–25 S	Later entries	Later profit
Put pair · 50%	0.437	1.007	151	$0.205m
Put pair · 75%	0.449	0.944	104	$0.195m
Put pair · 100%	0.478	1.055	75	$0.201m
Put pair · Learned sizing	0.573	1.263	110	$0.249m

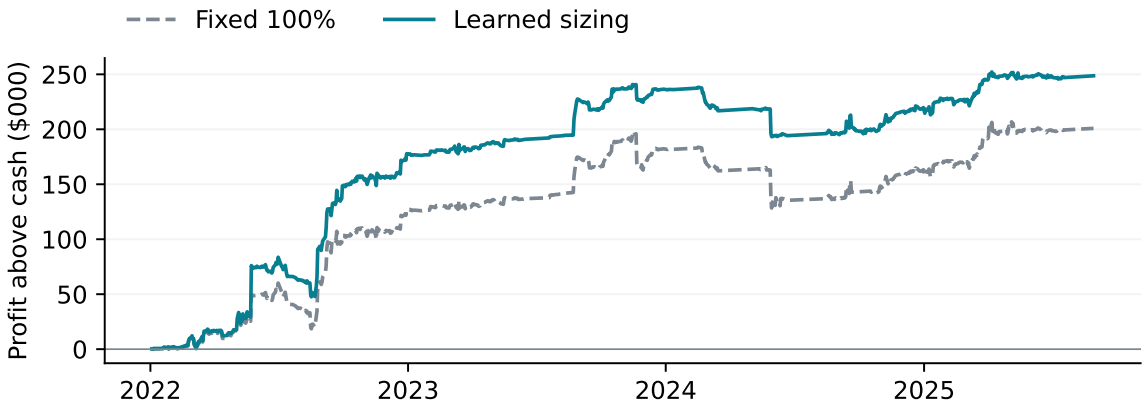


Figure 4. Put-pair profit above cash since January 2022 under learned and fixed-full entry sizing. Series subtract their 2021 closing values and retain carried positions. Differences reflect both sizing and admission.

The tree/CEM sizing policy improves later excess Sharpe by 0.208 and profit above cash by approximately $47,800 relative to full size. It admits 110 entries instead of 75.

As specified in Section 3.4, all policies receive the same 608 put-pair proposals, but size-dependent rounding, inventory and admission produce different funded entries. These comparisons measure complete sizing and admission policies rather than predictive sizing alone.

Most of the put learner's later lead develops in 2022, while 2024 loses ground. Its annualized total account return is 4.667%; growth relative to cash is 0.530%. Average reserve is 2.44% and peak reserve 15.77%. Cash interest explains much total growth; trading value is measured above cash.

The paired 95% bootstrap interval for the Sharpe gain of 0.208 is [-0.103, +0.424], conditional on the retained policies and without research-selection adjustment.

SNAP/UBER, GAP/TJX and AMD/MU supply 66.9% of later net pair trading profit, and 2022 supplies 83.1%, excluding shared financing. Sections C.1 and C.2 report annual and alternative-sizing results.

4.4.2 Policy Interpretation and Trade Admission

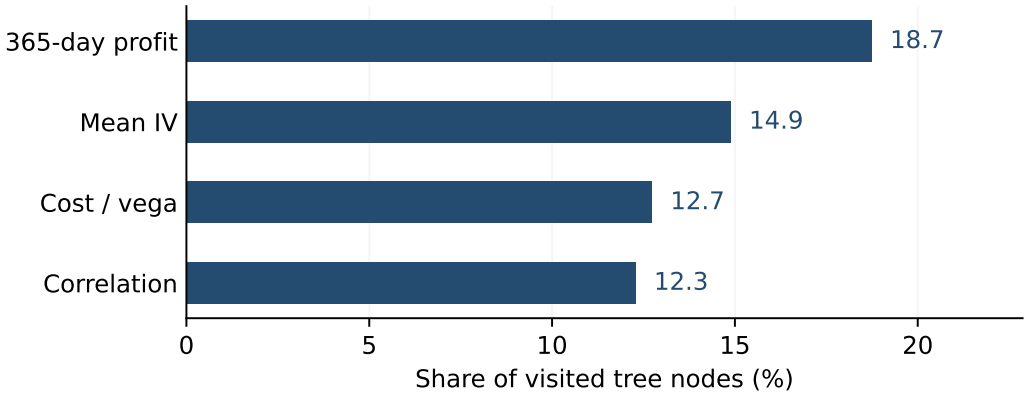


Figure 5. Put-pair policy usage during later evaluation: the share of visited tree nodes using each selected input. Frequencies describe fitted policy usage, not causal effects or profit contributions.

Put-policy decisions are reconstructed from the saved tree's root-to-leaf conditions and context-to-size map. Exact replay reproduces every saved size across all 608 put proposals. Completed 365-day profitability appears in 18.7% of later visited decision nodes, mean IV in 14.9%, commission relative to vega in 12.7%, and stock-return correlation in 12.3%. These frequencies describe which information the fitted policy uses; they are not causal effects or a ranking of expected profit contributions.

Later entry overlap with the fixed-full portfolio comprises 74 entries taken by both policies, 36 taken only by learned sizing and one taken only by fixed full size. These differences can arise through the inventory left by earlier size decisions. An explanation of proposed size alone therefore does not explain the entire portfolio result.

Joint sizing-and-allocation results follow in Section 6.4.

5 ATM Straddle Strategy

The straddle strategy sells approximately 30-day ATM straddles on underlyings with the lowest completed realized-minus-implied volatility (RV–IV) memories. The signal follows the volatility-premium motivation of Goyal and Saretto [5]; our experiments evaluate entry sizing and capital allocation around this fixed trading rule.

5.1 Trading Rules and Position Construction

For every historical candidate, subtract entry ATM-call IV from the annualized RV measured after the next-session entry close through that contract's expiry. Only after the outcome finishes can it join the signal history. Average those completed RV - IV observations over the preceding 365 calendar days, requiring 126. Rank eligible names each day and sell the bottom 10%.

$$M_{i,t} = \text{mean}_{u:\, t-365<T_u<t} \left(RV_{i,u\to T_u} - IV_{i,u}^{\text{ATM call}} \right).$$

Selection uses the cross-sectional bottom decile of eligible completed-outcome memories, without an absolute volatility-gap threshold.

Each position sells an ATM call and put with the same underlying, strike and expiry, with daily stock-delta rebalancing. Full size targets $450 combined call-and-put vega per IV point. The portfolio retains net short-gamma and short-vega exposure, including jump and volatility-revaluation risk.

The ranking uses ATM-call IV and annualized RV in volatility units, while execution uses the complete straddle. Profit is measured from the option-and-hedge ledger after costs and financing; the RV–IV ranking is neither a variance difference nor a profit identity.

5.2 Ridge-Based Contexts

The straddle sizing model fits a 60-month ridge model to completed full-size profit, using 14 inputs and requiring 128 examples. The inputs include IV, RV-IV memory, VIX and a separate fixed-half paper portfolio's recent performance. Score terciles define three contexts. Training uses only completed straddle outcomes and does not use the Q-allocation reward.

Training examples combine predetermined date-and-rank sampling of proposals with additional proposals admitted by a separate fixed-half benchmark portfolio. Only completed outcomes with finite reward labels enter fitting: 4,116 rewards are available in the saved sample. The fitted sizing policy is applied to all 35,774 proposals; future payoff availability does not restrict live proposals.

5.3 Position-Size Selection

Each context receives 50%, 75% or 100% exposure. We enumerate all 27 context-to-size maps and retain the highest Sharpe on their aggregate completed daily straddle-profit paths, including overlapping positions. Insufficient history uses half size. The context-to-size map need not increase monotonically with the ridge score because selection maximizes completed-path Sharpe. Section B.2 provides exact fitting details. Feature-usage and admission diagnostics follow in Section 5.4.

5.4 Results

5.4.1 Performance

Table 2. Straddle entry-sizing policies in validation and later evaluation. S denotes annualized excess Sharpe; later profit is profit above cash.

Entry policy	2018–21 S	2022–25 S	Later entries	Later profit
Straddle · 50%	0.740	0.920	1782	$1.824m
Straddle · 75%	0.542	0.968	1222	$2.010m
Straddle · 100%	0.528	0.990	940	$2.206m
Straddle · Learned sizing	0.775	1.041	1204	$2.379m

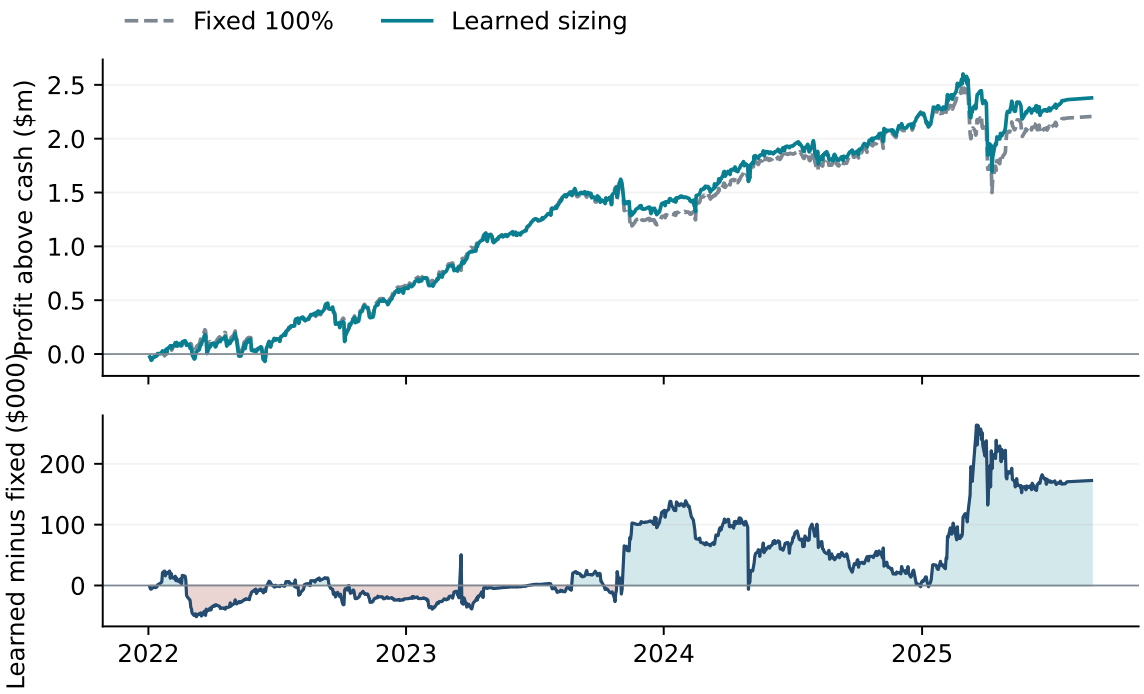


Figure 6. Straddle profit above cash since January 2022. Top: learned and fixed-full entry sizing. Bottom: incremental profit, learned sizing minus fixed full sizing. Series subtract their 2021 closing values and retain carried positions. Differences reflect both sizing and admission.

The learned straddle sizing policy reaches Sharpe 1.041 and $2.379 million above cash, compared with 0.990 and $2.206 million at full size. Its earlier Sharpe also exceeds each fixed-size control.

As specified in Section 3.4, all policies receive the same 35,774 straddle proposals, but size-dependent rounding, inventory and admission produce different funded entries. These comparisons measure complete sizing and admission policies rather than predictive sizing alone.

5.4.2 Policy Interpretation and Trade Admission

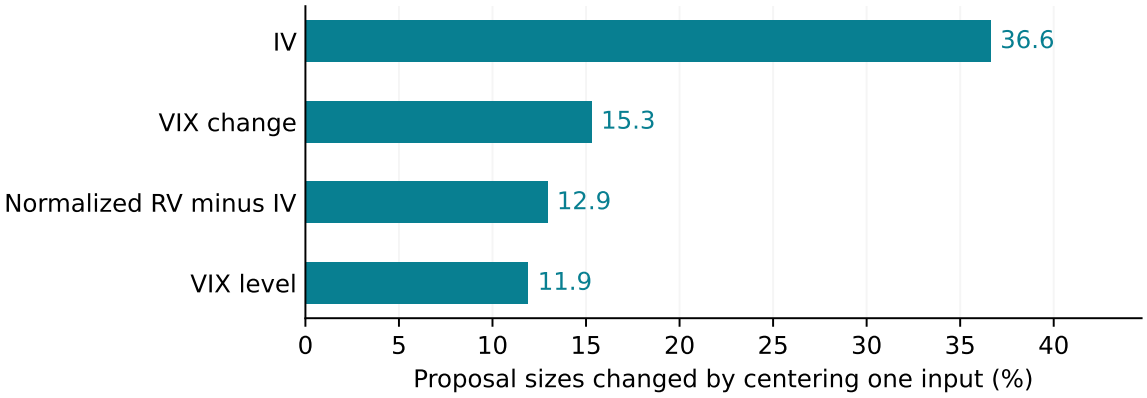


Figure 7. Straddle sizing sensitivity during later evaluation: the percentage of fitted proposal sizes that change when one input is centered. This descriptive sensitivity measure does not establish a causal effect on profit.

For a straddle proposal, standardized ridge contributions add to a score. Saved thresholds locate its context, and the learned map selects a size. Centering IV changes 36.6% of later fitted proposal sizes. Recent VIX change changes 15.3%, normalized RV-minus-IV memory 12.9%, and VIX level 11.9%. The size map need not rise monotonically with the regression score because it optimizes the combined daily path. Exact replay reproduces every saved size across all 35,774 straddle proposals.

Later entry overlap with the fixed-full portfolio comprises 897 entries taken by both policies, 307 taken only by learned sizing and 43 taken only by fixed full size. These differences can arise through the inventory left by earlier size decisions. An explanation of proposed size alone therefore does not explain the entire portfolio result.

Among quote-available completed straddle labels, full-size profit averages $1,631 for proposals assigned full size and $16 for those assigned half size. This deliberately sampled subset also reflects admission by the fixed-half benchmark and finite-label availability; it is not a representative sample of all proposals. Three within-date size shuffles yield later Sharpes 0.834, 0.928 and 1.044, versus 1.041 learned. One shuffle exceeds the learned policy, so the funded results and feature-usage diagnostics alone do not establish predictive sizing.

Completed-cohort attribution is consistent with collecting a premium for movement risk. For both more persistent and more reversing price paths, aggregate option-plus-hedge profit is positive when entry straddle IV exceeds subsequent RV and negative when it does not. These groups use realized outcomes and are not available entry screens. There are 220 losing cohorts even when entry IV exceeds subsequent RV. Option and stock-hedge profits must be read together; positive hedge profit is not an independent trading edge. Appendix D retains the full-history attribution, IV–RV definitions and accounting reconciliation.

Joint sizing-and-allocation results follow in Section 6.4.

6 Capital Allocation with Fitted Q-Learning

6.1 Allocation Framework and Execution

The put-pair portfolio offsets initial vega across names, whereas the straddle portfolio retains directional volatility exposure. Capital allocation determines the portfolio-level mix of these risks; entry sizing determines commitments within each strategy.

The fitted Q policy allocates one $10 million budget across the funded put-pair and straddle indices. The entry policies in Sections 4.2–4.3 and 5.2–5.3 construct these source portfolios independently: Q changes their capital weights rather than the tree, CEM size map or straddle ridge score. The allocation model is a separate linear-in-parameters value function fitted through Bellman regression.

The seven put/straddle capital targets are 100/0, 90/10, 75/25, 50/50, 25/75, 10/90 and 0/100, without outer leverage or a cash action. Weekly decisions use prior-close information and execute at the next close: the first session retains existing holdings, followed by resizing and weight drift until the next decision.

6.2 State and Action Representation

The state contains nine inputs, defined in Table 3. The first four describe volatility opportunities; the next four describe source-portfolio risk and recent performance. The final input is the drifted straddle allocation before the next decision, which matters because allocation changes incur costs. Actions are fractional claims on funded strategy indices rather than a shared integer-contract order book. Source returns already include underlying trading costs; resizing adds source-based option and hedge charges.

All inputs use information available at the close immediately preceding a weekly decision. Table 3 describes their construction; the shared numerical conventions below preserve the exact calculations.

Table 3. The nine allocation-state inputs. All inputs are dimensionless and use information available through the prior close.

State input	Calculation
Put-pair spread-rank strength	Average $2\|q - 0.5\|$ across pair proposals, where q is each pair's selected-put IV-spread percentile over 252 sessions, requiring 63 observations.
Put-pair relative variance gap	Average the direction-adjusted, normalized difference between the two legs' implied-minus-realized variance gaps. The equation below defines each pair's contribution.
Straddle IV–RV richness	Subtract trailing 21-session RV from ATM-call IV and divide by that IV. Average across straddle proposals, then smooth over 21 sessions.
Completed straddle volatility premium	Negate the completed RV–IV memory defined in Section 5 and divide by current ATM-call IV. Average across straddle proposals, then smooth over 21 sessions. This is a volatility signal, not cash premium received or trading profit.
Relative strategy volatility	Natural log of straddle volatility divided by put-pair volatility, using the funded indices' trailing daily excess returns.
Strategy-return correlation	Trailing Pearson correlation between the two funded indices' daily excess returns.
VIX level	Latest available closing VIX level divided by 20.
Relative recent Sharpe	Straddle Sharpe minus put-pair Sharpe over the same trailing window, capped between -5 and 5.
Previous straddle allocation	The straddle's portfolio weight after returns have caused it to drift from the previous target. Start at 50/50; subsequently multiply each previous target weight by its index's cumulative gross return since execution, then divide by their combined value.

Aggregation and lookbacks Proposal averages are equally weighted. On dates without new proposals, carry forward the most recent cross-sectional average. For the two straddle opportunity inputs, apply the 21-session moving average after this carry-forward step, requiring at least one observation. For their denominators, floor ATM-call IV at 0.01 in decimal units. Straddle trailing RV is the square root of 252 times the mean of 21 squared adjusted-stock log returns. The completed RV–IV memory uses a 365-calendar-day window and requires 126 completed observations.

For the put-pair variance-gap input, each proposal contributes

$$d\,\frac{(I_A^2 - v_A) - (I_B^2 - v_B)}{[(I_A + I_B)/2]^2}.$$

Here I_A and I_B are the selected-put IVs, and v_A and v_B are annualized stock variances: 252 times the trailing 20-session mean of squared simple returns, requiring ten observations. Direction $d = +1$ means buy A and sell B; $d = -1$ reverses the position. An unavailable proposal-level variance gap is set to zero before averaging. This normalized, relative gap is not an absolute variance-risk-premium estimate.

Risk inputs and numerical conventions Strategy volatility, correlation and Sharpe use up to 63 daily excess-return observations, requiring at least 21. Volatility is the sample standard deviation. Annualized Sharpe is the mean daily excess return divided by its standard deviation, multiplied by the square root of 252. In the volatility ratio, floor each standard deviation at 10^{-8}. Set an

unavailable strategy Sharpe or correlation to zero before forming the relative-Sharpe input; set other unavailable or nonfinite state values to zero. Recompute risk and Sharpe inputs for each entry-policy family, while retaining the same economic opportunity definitions. The previous-allocation input uses drift through the prior close; execution remains at the next close.

Let a denote the target straddle weight and let $x(s)$ contain an intercept and the nine standardized state inputs. The value function is

$$Q_\theta(s,a) = \beta_0^\top x(s) + a\,\beta_1^\top x(s) + a^2\beta_2^\top x(s), \qquad a \in \{0, .10, .25, .50, .75, .90, 1\}.$$

Each coefficient block has ten parameters, giving 30 in total. The model is linear in its fitted coefficients and quadratic in the allocation action; the state-action interactions allow the preferred weight to change with market conditions. Training-window means and standard deviations standardize the state, with a scale floor of 0.1 and clipping within ± 5.

6.3 Reward and Fitted Q-Learning

The weekly reward balances return above cash against the size of daily fluctuations. First, divide each daily excess return by a common reference volatility, estimated from the training window's fixed-50/50 portfolio. This gives all candidate allocations the same reward scale. Then subtract a penalty proportional to the squared normalized return and sum over the week:

$$R_w(\lambda) = \sum_{d \in w} (z_d - \lambda z_d^2), \qquad z_d = e_d / \widehat{\sigma}_{\text{ref}}.$$

Here e_d is the allocated portfolio's daily excess return, including source trading and outer resizing costs. The parameter λ controls the fluctuation penalty: larger values make large daily moves more costly. Because the return is squared, the penalty applies to both gains and losses, not only downside returns. At $\lambda = 0$, the objective is expected discounted cumulative normalized excess returns without an explicit fluctuation penalty. It does not directly maximize compounded dollar profit or Sharpe.

The learner first fits the 30-parameter Q-function to immediate weekly rewards by ridge regression. It then repeatedly updates the regression target to include the best estimated future value from the following state, using fitted Q iteration [6]. For a completed transition (s_w, a_w, R_w, s_{w+1}), each subsequent pass uses

$$y_w^{(k)} = R_w(\lambda) + \gamma \max_{a' \in \mathcal{A}} Q_{\theta^{(k-1)}}(s_{w+1}, a'),$$

The model refits the same regression basis to these updated targets. The first term rewards the current week's allocation; the second accounts for continuation value, including the effect of inherited holdings on future resizing costs. The discount $\gamma = 0.99$ weights that future value and remains fixed across risk preferences. The matched $\gamma = 0$ control uses the same regression architecture but removes the continuation term.

Each monthly fit uses the preceding 104 completed weeks, requiring at least 52. For each historical market week, we replay the same source-strategy returns under seven inherited allocations and seven new capital targets, producing 49 counterfactual paths that account for weight drift and resizing costs. Each week receives total regression weight one; these paths are not independent market observations. Ridge regularization uses coefficient 10, except for the global intercept, which receives a penalty of 0.01. There are 200 regression passes in total: one fit to immediate rewards followed by 199 continuation-value updates. Section B.2 reports the convergence checks.

Between monthly fits, the policy scores all seven capital splits at each weekly decision and selects the highest estimated Q-value. It uses 50/50 before enough training weeks exist; each fixed control starts at its own target. In the four-way comparison, Q is independently refitted for each entry-policy family. Source returns, strategy-dependent risk features, rewards and resizing charges are rebuilt for that family, while economic signal definitions remain shared. This downstream refitting does not jointly retrain either source sizing policy.

6.4 Results

Standalone entry-sizing results are reported in Sections 4.4 and 5.4. We evaluate their interaction with capital allocation and conventional dynamic benchmarks.

6.4.1 Joint Sizing and Allocation

Table 4. Separating learned entry sizing and learned capital allocation. S denotes annualized excess Sharpe; Q uses the illustrative risk preference $\lambda = 0.20$, not the original validation-selected setting. Fixed entries use full size and fixed allocation is 50/50. Q is independently fitted for each entry-policy family.

Entry / capital policy	2018–21 S	2022–25 S	Later profit
Fixed / fixed	0.556	1.088	\$1.145m
Learned sizing / fixed	0.804	1.167	\$1.238m
Fixed / Q	0.870	1.194	\$0.955m
Learned sizing / Q	1.215	1.726	\$1.306m

At this illustrative risk preference, both learning layers improve Sharpe in both periods against full-size entries and 50/50 allocation, with the largest point-estimate gain when combined. The original validation-selected Q policy uses $\lambda = 0.025$, with validation Sharpe 1.437 and later Sharpe 0.282 (Table 7). Thus the four-way comparison isolates the two decisions at a stated risk preference; it does not show that the validation-based selection rule transferred successfully. Q is independently refitted on each entry family's completed returns, risk features and costs, rather than transferring the policy fitted to learned-sizing portfolios.

Table 5. Allocation with constant 75% entry sizing.

Fixed-75% entries	2018–21 S	2022–25 S	Later profit	Drawdown
Fixed 50/50	0.566	1.070	\$1.053m	3.16%
Q, $\lambda = .20$	0.891	2.118	\$1.060m	0.43%

The fixed-75% result shows that capital allocation can help independently of entry learning. Q nearly doubles later Sharpe with similar profit and a much smaller drawdown. Its later Sharpe exceeds the combination of learned sizing and Q at this λ, while its profit and validation Sharpe are lower. Learned sizing therefore changes the available profit–risk tradeoff without dominating every constant entry size.

6.4.2 Dynamic Allocation Benchmarks

We benchmark Q against minimum-variance, maximum-Sharpe and mean–variance allocation, adapting the objectives of Caner et al. [7] to the two frozen learned-sizing portfolios. These controls distinguish the Q result from gains available through conventional dynamic allocation.

Experimental Controls All methods choose among the same seven capital splits, use one $10 million budget and pay the same source trading and additional resizing costs. Estimates use Q's preceding 104 completed weeks, with a minimum of 52, monthly refits and the same 50/50 warmup. Within each objective, 2018–2021 Sharpe selects sample covariance or linear Ledoit–Wolf shrinkage; that choice remains fixed for the later period. This adapts their objectives rather than replicating their stock screens, nonlinear shrinkage or 15-year estimation window.

Table 6. Q-learning and conventional dynamic allocation benchmarks.

Capital allocator	2018–21 S	2022–25 S	Later profit	Later σ (%)
Fixed 50/50	0.804	1.167	$1.238m	1.973
Minimum variance	0.316	1.329	$0.284m	0.438
Maximum Sharpe	0.189	0.659	$0.353m	1.154
Mean–variance, $\lambda = .20$	0.355	1.519	$0.452m	0.619
Q, $\lambda = .20$	1.215	1.726	$1.306m	1.353
Mean–variance, $\lambda = 1$	0.246	1.795	$0.439m	0.519
Q, $\lambda = 1$	1.005	2.152	$1.141m	0.985

All S columns are annualized excess Sharpe; later σ is annualized daily excess-return volatility. At $\lambda = 0.20$, Q raises later Sharpe from 1.519 to 1.726 and profit above cash from $452,303 to $1,305,940. At $\lambda = 1$, the corresponding figures are 1.795 versus 2.152 and $439,118 versus $1,141,251. Q also improves both measures at $\lambda = 0.10$ in both periods. These are portfolio comparisons, not equal-risk experiments: Q takes more realized risk.

Allocation Objectives and Policy Differences The conventional rules summarize each trailing window with one mean and covariance estimate. Q also conditions on current volatility pricing, completed premium, VIX and relative strategy performance. At $\lambda = 0.20$, its mean later straddle target is 24.9%, versus 12.7% for mean–variance allocation. The selected intervals in Section 6.4.4 have higher subsequent straddle returns and lower volatility, supporting selective participation rather than average exposure alone.

Q changes targets 24 times later, versus eight for mean–variance allocation. Removing only outer resizing charges raises their Sharpes to 1.774 and 1.525; all source trading costs remain. Lower resizing fees therefore do not explain Q's advantage. The conventional rule uses centered variance, whereas Q penalizes the raw second moment and adds state and continuation value; identical λ values do not make these identical objectives.

The result has a clear exception: at $\lambda = 0$, conventional profit-only allocation reaches later Sharpe 1.041 and $2.154 million above cash, versus Q's 0.671 and $1.353 million. The literal 1% monthly target used in [7] is infeasible under our allocation limits; its maximum-mean fallback is recorded separately. All 14 conventional cases and the fee diagnostics are retained in the companion results.

6.4.3 Risk Preferences

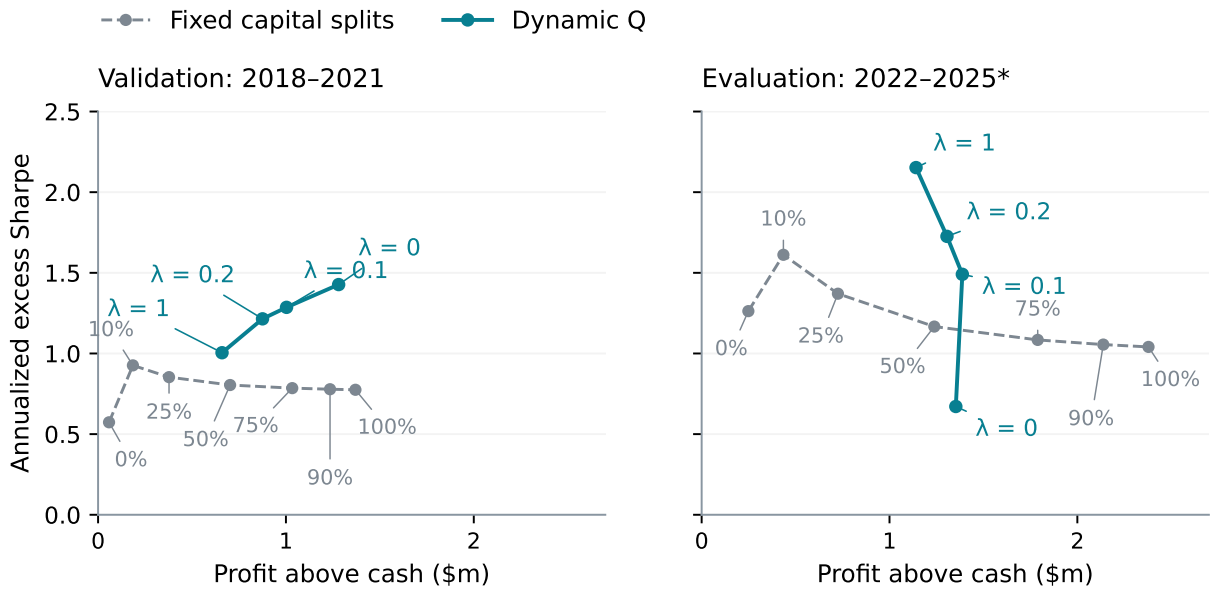


Figure 8. Four illustrative Q preferences on learned-sizing source portfolios; the original validation-selected policy is reported in Table 7. Nine-input Q state and $\gamma = 0.99$. Gray labels are fixed straddle percentages; the remainder goes to put pairs. Lines connect discrete policies. Both axes have the same scales across periods; 2025 ends in August.

Table 7. Original validation-selected Q policy and four illustrative reward preferences. All use the same learned-sizing source portfolios, capital budget and execution assumptions, with $\gamma = 0.99$.

Reward	2018–21 S	2018–21 profit	2022–25 S	2022–25 profit
$\lambda = .025$ (selected)	1.437	\$1.244m	0.282	\$0.611m
$\lambda = 0$	1.427	\$1.281m	0.671	\$1.353m
$\lambda = 0.1$	1.286	\$1.003m	1.491	\$1.388m
$\lambda = 0.2$	1.215	\$0.875m	1.726	\$1.306m
$\lambda = 1$	1.005	\$0.660m	2.152	\$1.141m

The original selection rule maximizes 2018–2021 validation Sharpe within each discount setting and retains that choice for evaluation. At $\gamma = 0.99$, it selects $\lambda = 0.025$; the later Sharpe falls to 0.282 despite positive profit above cash. The four illustrative preferences describe sensitivity to the objective and do not replace that selection record. Reward preferences change the profit–risk tradeoff. At $\lambda = 0.20$, Q's later Sharpe of 1.726 exceeds the highest-Sharpe fixed split, 90/10, at 1.612; their profits above cash are \$1.306 million and \$435,181. Increasing λ to 1 raises Q's Sharpe to 2.152 while reducing profit, and straddle-heavy fixed allocations still generate more dollars. Annualized cash-relative growth and excess volatility are 2.34% and 1.35% at $\lambda = 0.20$, versus 2.13% and 0.98% at $\lambda = 1$.

At $\lambda = 0$, the learner aggressively pursues profit through the higher-dollar, higher-variance straddle strategy. It chooses straddles alone in 138 of 191 later decisions, with a mean straddle target of 73.2%, versus 24.9% at $\lambda = 0.20$ and 16.9% at $\lambda = 1$. Validation Sharpe is 1.427, but later Sharpe falls to 0.671. When straddle performance weakens, this concentration becomes costly: the zero-penalty portfolio loses approximately $331,000 above cash in 2025.

The problem also involves when capital is committed. A fixed 75% allocation to straddles achieves later Sharpe 1.084, so average exposure alone does not explain the weaker Q result. The profit-only learner can respond to state, but its learned timing does not distinguish attractive straddle opportunities well enough in the later period. Positive penalties make large fluctuations costly and lead to more selective straddle exposure. Increasing Bellman updates from 200 to 500, 1,000 and 2,000 changes no later decision for these settings. This check points to the reward and learned allocation rule, rather than too few iterations.

6.4.4 Policy Interpretation

Straddles offer more dollar profit and much greater daily risk. At $\lambda = 0.20$, Q chooses put pairs alone on 141 later decision dates, straddles alone on 46, and an interior mix on four. It primarily switches strategies.

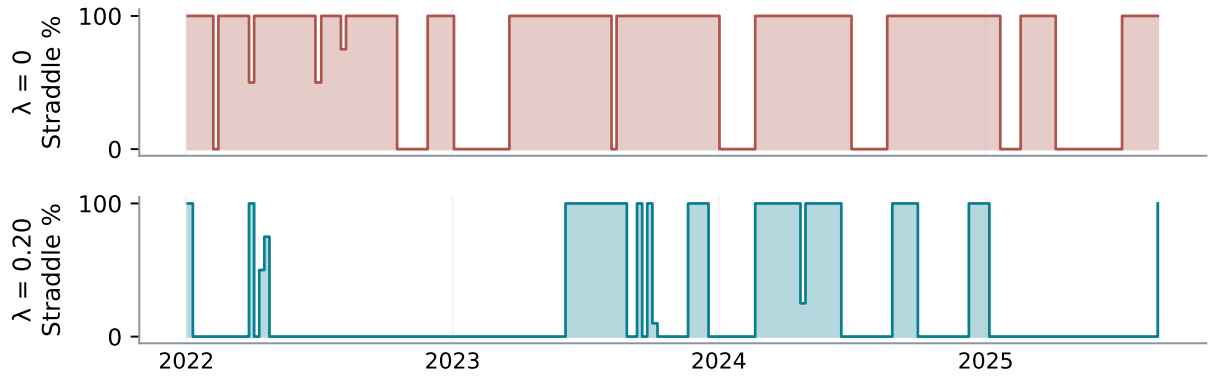


Figure 9. Weekly target capital allocation to the straddle strategy during evaluation: top, $\lambda = 0$; bottom, $\lambda = 0.20$. Shaded height shows the straddle percentage; the remainder is allocated to put pairs.

Conditional Strategy Returns and Risk After Q chooses straddles, their mean daily excess return is 3.05 basis points with volatility 16.18; after it chooses put pairs, the straddle figures are 1.30 and 27.00. Put pairs average -0.09 basis points on the first set and +0.31 on the second. These are realized conditional statistics on noncontiguous dates, not causal estimates or standalone strategy Sharpes. Source costs are included; outer resizing charges are excluded.

A fixed target equal to Q's realized mean allocation reaches later Sharpe 1.387, versus 1.726 for Q: average exposure alone does not reproduce the result. The matched discount-zero learner reaches 1.645; continuation changes nine of 191 decisions at the same states. Much of the gain comes from conditional allocation, with a smaller incremental sequential benefit.

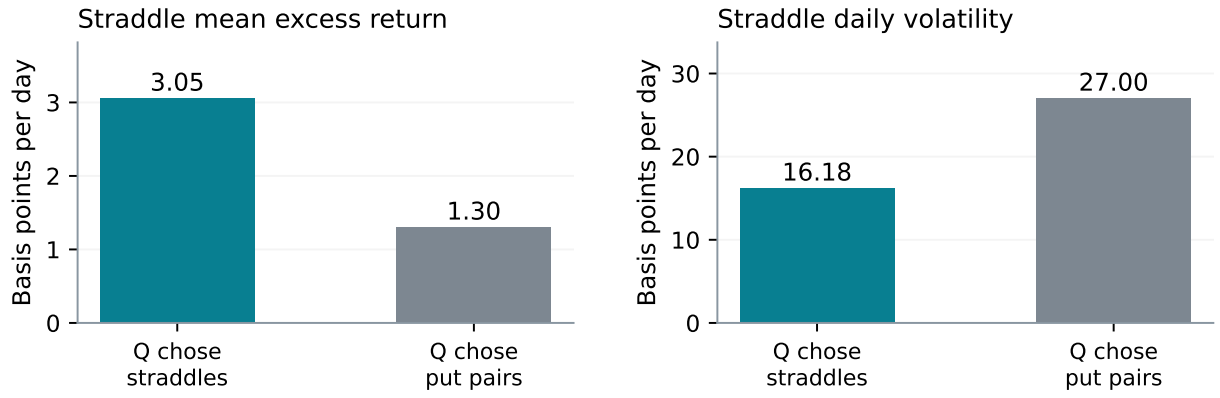


Figure 10. Realized straddle returns after $\lambda = 0.20$ Q decisions. Straddle-only choices cover 220 sessions; put-only choices cover 678. The four mixed-allocation intervals are omitted. Source costs are included; outer resizing charges are excluded.

7 Discussion and Conclusion

Summary of Findings

Learned entry policies improve the retained portfolios through sizing and admission. Q improves allocation with constant entries and exceeds the corresponding rolling mean–variance controls at $\lambda = 0.10$, 0.20 and 1. Its profit-only policy transfers less reliably when straddles weaken. These are illustrative risk-preference comparisons. The original validation-selected $\lambda = 0.025$ policy falls from validation Sharpe 1.437 to later Sharpe 0.282, so validation-based policy selection did not transfer its strong historical performance. Reward design changes the allocation tradeoff.

Limitations

All fitted policies use only information available before their decisions. However, the extent to which later-period inspection influenced development, including training-window and risk-preference choices, remains to be verified (Section B.1). The evaluation should therefore be interpreted as historical performance under chronologically fitted policies, rather than a confirmed independently frozen test of the complete research process. The controlled comparisons provide evidence about adaptive exposure allocation under the stated assumptions, but neither the magnitude of research-selection bias nor prospective performance has been established.

Execution is another boundary. Midpoint-based fills with an adverse charge do not establish tradability, particularly for paired orders that may fill asynchronously. Quoted-spread execution, liquidity capacity, pin risk, early assignment and forced liquidation need further work. The outer experiment reallocates fractional strategy indices, not shared integer contracts. Initial vega matching is local, and the straddle signal adds net short-volatility risk.

Conclusion

The experiments support adaptive exposure allocation across portfolios with different capital deployment and volatility risks. The four-way comparison separates entry learning from allocation, while the dynamic benchmarks and one-period control place the Q result in context. The evidence is strongest for conditional allocation under the stated historical simulation assumptions; policy rankings depend on the profit–risk objective, and the original validation-selected policy performs weakly in the later period. The headline illustrative policy therefore does not establish successful validation-based selection or prospective performance.

Appendix D relates straddle profitability to realized movement. An integrated implementation should combine entry sizing, allocation and inventory controls under measured execution costs, followed by prospective evaluation and tests on additional independent signals.

References

[1] B. Ning and K. Lee. Advanced statistical arbitrage with reinforcement learning, 2024. URL `https://arxiv.org/abs/2403.12180`.

[2] Sebastian Bell, Ali Kakhbod, Martin Lettau, and Abdolreza Nazemi. AlphaGlass: Interpretable characteristic-based portfolio choice. Working Paper 35186, National Bureau of Economic Research, May 2026. URL `https://www.nber.org/papers/w35186`.

[3] Igor Halperin. QLBS: Q-learner in the Black-Scholes(-Merton) worlds, 2017. URL `https://arxiv.org/abs/1712.04609`.

[4] Pieter-Tjerk de Boer, Dirk P. Kroese, Shie Mannor, and Reuven Y. Rubinstein. A tutorial on the cross-entropy method. *Annals of Operations Research*, 134:19–67, 2005. doi: 10.1007/s10479-005-5724-z. URL `https://people.smp.uq.edu.au/DirkKroese/ps/aortut.pdf`.

[5] Amit Goyal and Alessio Saretto. Cross-section of option returns and volatility. *Journal of Financial Economics*, 94:310–326, 2009. doi: 10.1016/j.jfineco.2009.01.001.

[6] Damien Ernst, Pierre Geurts, and Louis Wehenkel. Tree-based batch mode reinforcement learning. *Journal of Machine Learning Research*, 6:503–556, 2005. URL `https://jmlr.org/papers/v6/ernst05a.html`.

[7] M. Caner, A. Capponi, N. Sun, and J. Y. Tan. Designing agentic AI-based screening for portfolio investment, 2026. URL `https://arxiv.org/abs/2603.23300`.

A Contract Selection, Features, and Risk Constraints

This appendix separates contract selection, the inputs to each strategy's sizing model, and the admission rules used in the funded contract-level replays. Allocation-state inputs are defined separately in Section 6.

A.1 Data and Contract Selection

The data universe and coverage limitations are described in Section 3. Table 8 distinguishes the two contract screens.

Table 8. Strategy-specific contract selection.

Criterion	Put pair	Straddle
Expiry	Nearest 30 days; entry requires 7–55 days remaining	ATM-call signal screen requires 21–45 days from next-session entry to expiry
Strike or delta	Put nearest absolute delta 0.25, within 0.075	ATM-call strike within 10% of spot
Position legs	Selected put on each related underlying	Call and put match on underlying, strike, expiry and adjustment

Both screens require standard 100-share contracts, positive noncrossed quotes, open interest of at least 100 and quote width no more than 25% of midpoint. Put-selection ties prefer narrower quotes, higher open interest and then lower contract identifier.

Quote handling and valuation Missing entry quotes reject proposals. Corporate events can force exits. Held marks may use last-known IV with current spot, rates and known dividends. European-style valuation and cash settlement approximate expiry; assignment is not fully simulated.

A.2 Put-Pair Sizing Inputs

The tree uses 15 numerical inputs and 12 binary availability flags, giving 27 inputs in total. In Table 9, starred inputs are required; every other numerical input has an availability flag.

Table 9. Numerical inputs to put-pair sizing; starred inputs are required.

Input	Definition
Signal and volatility	
Mean IV*	Mean implied volatility of the selected puts
Directional IV gap*	IV gap oriented to the proposed trade direction
Spread-rank strength*	Strength of the historical IV-spread rank signal
Directional variance gap	Oriented IV-squared minus trailing-variance gap
Mean realized volatility	Mean trailing stock realized volatility
Pair relatedness	
Return beta	126-session stock-return beta
Return correlation	126-session stock-return correlation

Continued on next page

Put-pair sizing inputs (continued)

Input	Definition
Contract terms and costs	
Mean time to expiry	Mean days remaining to expiry
Directional expiry gap	Expiry difference oriented to trade direction
Commission relative to vega	Training commission expressed in IV points
Market conditions	
VIX	Strictly prior VIX close
Completed directional outcomes	
91-day profitability	Mean completed profit per initial vega over 91 calendar days
91-day observation count	Number of completed observations in that window
365-day profitability	Mean completed profit per initial vega over 365 calendar days
365-day observation count	Number of completed observations in that window

An availability flag is one when its numerical field is observed and zero otherwise. Missing optional values are set to zero with flag zero; an observed numerical zero retains flag one. Put trailing variance is 252 times mean squared simple stock returns over 20 sessions, requiring at least ten observations.

A.3 Straddle Sizing Inputs

The ridge model uses the 14 inputs in Table 10. Benchmark features come from a separate fixed-half straddle portfolio constructed using information available at each date, not the learner's future inventory.

Table 10. All 14 inputs to the retained straddle ridge model.

Input	Definition
Volatility opportunity and recent changes	
Completed RV–IV memory	Historical completed realized-minus-implied volatility memory from Section 5
IV	Current ATM-call implied volatility
VIX	VIX level available at the proposal date
Normalized memory	Negative RV–IV memory divided by current IV, with IV floored at 0.01
Variance ratio	Forecast annualized variance divided by squared IV
21-session VIX change	Proportional change in VIX over 21 sessions
5-session IV change	Proportional change in IV over five sessions
Fixed-half benchmark portfolio	
21-session Sharpe	Annualized excess Sharpe over 21 sessions
63-session Sharpe	Annualized excess Sharpe over 63 sessions
126-session Sharpe	Annualized excess Sharpe over 126 sessions
63-session volatility	Annualized daily excess-return volatility over 63 sessions

Continued on next page

Straddle sizing inputs (continued)

Input	Definition
63-session drawdown	Cash-relative wealth divided by its trailing 63-session maximum, minus one
Reserve use	Capital reserve divided by current portfolio NAV
Benchmark availability	Binary flag indicating that the benchmark features are available

A.4 Risk Limits and Admission

The limits in Table 11 govern funded contract-level admission, not the outer strategy-index allocation.

Table 11. Risk limits used in contract-level admission checks.

Check	Limit or definition
Stress loss	At most 5% of the applicable budget
Capital reserve	At most 80% of the applicable budget
Hedge notional	At most twice the applicable budget
Gross vega	At most \$20,000 per IV point per \$1 million of reference capital
Reserve measurement	Gross option premium plus half the netted hedge notional
Budget scope	Global checks use equity; pair checks use a \$1 million reference scaled by current divided by initial equity

Stress testing takes each name's worst modeled option-plus-hedge loss across spot shocks of $-20\%, 0, +20\%$ and IV shocks of $-10, 0, +10$ points. These losses are summed without pair-correlation credit. Quote eligibility, inventory and the limits above jointly determine whether a proposed size is admitted.

A.5 Data and Code Availability

Supporting materials, saved results and figures are available at `https://github.com/kreitzz/rl-atm-vega-pairs`. Access to the underlying WRDS data requires the relevant subscriptions and vendor permissions; redistribution restrictions apply. Model fitting requires the licensed source cache or suitable archived derived inputs.

Public snapshots support reproduction of the attribution and associated figures from saved derived data, without raw option chains. This verifies archived results rather than independently refitting the models or regenerating all trading experiments. Source locations, verification details and reproduction guidance are provided in the paper's repository README.

B Model Estimation and Selection

B.1 Model Selection and Research Chronology

The put-sizing specification was selected from 33 archived screened configurations ranked by 2018–2021 validation excess Sharpe within the established 30-day, 25-delta design and pair roster. The grid varied features, training windows and action menus. Baseline, IV-factor and term-structure scores tie exactly at 0.575; alphabetical configuration ordering retains the baseline. The selected policy uses 27 inputs, an 18-month window and 50/75/100% sizes. Selection used a $345 million research account; the funded comparisons use $10 million.

The straddle's 48- and 60-month windows tie at validation Sharpe 0.775; their later Sharpes are 0.960 and 1.041. **Verification pending:** an earlier draft attributes refinement to these later-period results. Confirm whether and when this occurred against experiment records and author notes; the reported scores alone do not establish the development chronology.

Table 7 reports four illustrative Q preferences alongside the original validation-selected $\lambda = 0.025$ policy at $\gamma = 0.99$. Its validation and later Sharpes are 1.437 and 0.282; the illustrative settings do not replace that selection. Supporting materials retain the original selection record and full results (Section A.5); chronological fitting does not establish an untouched evaluation.

B.2 Additional Estimation Settings and Convergence

Monthly put refits use only completed outcomes with signal dates in the preceding 18-month window; settings remain fixed while trees and size maps update. Table 12 supplements the main model descriptions. Future profits are training outcomes, never current inputs.

Table 12. Additional entry-policy estimation settings.

Component	Setting
Put CEM probability updates	Seed 2026; uniform initial probabilities. Mix 30% prior with 70% elite frequency, multiply by 0.97 and add 0.03 spread uniformly across actions.
Put training and evaluation costs	Option commission: $0.65 for memory and training labels; $0.03 for every funded policy. Training scores rounded-contract daily-profit paths; evaluation uses funded NAV returns.
Straddle ridge preprocessing	Median centers; standard deviations floored at 0.001; standardized inputs clipped within ±5.
Straddle fitting and map selection	Ridge penalty 100, with no intercept penalty; each proposal date receives equal total weight. Full-size completed profit is the target. Size-map ties favor lower exposure.

For Q convergence, all 81 monthly fits were checked at 200, 500, 1,000 and 2,000 regression passes. Later actions are identical across budgets for the four illustrative λ values. The Q-function basis, normalization, penalties and regression weights are specified in Section 6.

B.3 Statistical Inference

The paired interval for the put-sizing Sharpe difference uses 1,000 circular block-bootstrap resamples with 126-session blocks, holding the compared policies fixed. The interval conditions on the retained policies and historical paths; it does not adjust for model or research selection. The estimated gain and interval are reported in Section 4.4. Neither this interval nor chronological fitting measures the bias from development choices or establishes prospective performance.

C Supplementary Results and Diagnostics

The tables and diagnostics below supplement the main policy comparisons. Reporting measures are defined in Section 3.5.

C.1 Annual Performance and Reward Sensitivity

Table 13. Annualized excess Sharpe by calendar year; 2025 ends in August.

Year	**Put sizing**	**Straddle sizing**	**Q: $\lambda = .20$**	**Q: $\lambda = 1$**
2018	0.522	0.148	0.160	0.160
2019	-0.542	1.459	0.112	-0.237
2020	1.555	0.161	1.617	1.028
2021	-0.580	2.336	2.932	3.180
2022	2.458	1.206	2.042	2.540
2023	1.492	1.727	2.449	2.907
2024	-0.907	1.465	1.945	1.954
2025*	1.092	0.085	-0.598	1.092

All figures are annualized excess Sharpe; 2025 ends on 29 August. The sleeves retain their separate funded histories, and outer allocations include resizing charges. Annual results show substantial regime variation.

Table 14. Later excess Sharpe across entry policies and Q reward preferences.

Entry policy + Q	$\lambda = 0$	$\lambda = .10$	$\lambda = .20$	$\lambda = 1$
Fixed 50%	0.541	0.703	1.116	1.349
Fixed 75%	0.606	0.667	2.118	1.897
Fixed 100%	0.618	1.354	1.194	1.633
Learned sizing	0.671	1.491	1.726	2.152

This table holds Q's feature definitions, discount, training window and action menu fixed while refitting separately on each entry family. Values are 2022–2025 Sharpes. The complete seven fixed capital splits are also retained; 50/50 is the primary fixed-allocation baseline, rather than a later-period optimized control.

C.2 Additional Put-Sizing Results

Table 15. Alternative put-sizing models on the common reporting split. Fixed-size controls are reported in Section 4.4.

Put sizing policy	2018–21 S	2022–25 S	Later entries	Later profit
Tree + CEM	0.573	1.263	110	$0.249m
Spread-rank sizing	0.464	0.815	121	$0.165m
Ridge-score sizing	0.421	0.890	94	$0.180m
Direct linear policy	0.267	0.873	96	$0.171m
Direct additive policy	0.302	1.013	97	$0.180m

All five policies receive the same put proposals, directions, contracts and funded admission rules. The four alternatives to tree/CEM are follow-up comparisons, not original selection candidates. Tree/CEM has the highest funded Sharpe in both periods among these policies; this researched comparison does not establish superiority on an untouched test.

C.3 Allocation Timing Diagnostics

Table 16. Allocation timing controls at $\lambda = 0.20$.

Allocation control, $\lambda = .20$	2018–21 S	2022–25 S	Later profit
Fixed at Q's mean target	0.828	1.387	$0.841m
One-period learner, $\gamma = 0$	1.053	1.645	$1.184m
Sequential Q, $\gamma = 0.99$	1.215	1.726	$1.306m

The mean-weight control uses Q's realized average target retrospectively. 49 nonzero circular rotations preserve allocation counts and most serial structure, then recompute wealth and resizing costs. None reaches actual later Sharpe; three reach or exceed actual validation Sharpe, and median later Sharpe is 0.744. These are descriptive information-use and timing diagnostics, without causal feature attribution or research-selection adjustment.

C.4 Reporting Boundaries and Account Construction

Reporting periods retain the accounts' existing capital and open positions: accounts are neither reset nor liquidated at the 2021 split. Performance curves subtract each policy's excess dollar wealth at the preceding close. Account growth includes cash interest; excess returns and profit above cash use the benchmark defined in Section 3.5.

A final distinction concerns portfolio admission. The source put and straddle books each simulate whole contracts and daily stock hedges under their own funded histories. The outer allocator trades fractional claims on those net strategy indices, with additional resizing charges. Its capital weights are not premium deployment or broker margin, and reallocating indices does not rerun a common integer-contract inventory ledger.

D Straddle Attribution and Reconciliation

This appendix explains the straddle account's full-history profit, January 2018–August 2025. It supports the strategy's economic interpretation, not the incremental value of learned sizing or Q allocation.

D.1 Straddle Profit Attribution

Across 2,599 executed cohorts under learned sizing, options after option charges earned $1.74 million and stock hedges before hedge costs earned $1.75 million. Their combined $3.49 million reconciles to $3.75 million above cash after the remaining costs, dividends and financing, as shown below.

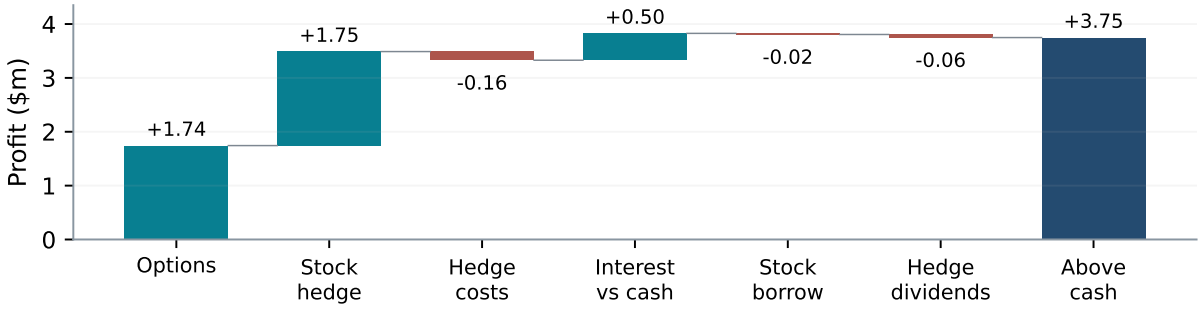


Figure 11. Full-history straddle account, January 2018–August 2025. The option and stock components are read together; the remaining cash flows reconcile them to profit above cash.

The option and hedge components must be read together. Option marks reflect time decay, stock-price moves and volatility revaluation; the stock position offsets option delta. Positive profits in both components therefore do not establish two independent sources of alpha.

D.2 Implied and Realized Volatility

Table 17 groups the same cohorts by entry IV versus subsequent RV and completed price path. Combined profit is positive in both path groups when IV exceeds RV and negative when it does not. This is consistent with compensation for selling movement risk. The groups use realized outcomes, so they are descriptive rather than available entry screens.

Table 17. Full-history straddle outcomes by completed path and IV versus subsequent RV. Profit combines options after option charges and stock hedges before hedge charges, dividends and financing; totals depend on cohort counts and sizes.

Price path	IV vs RV	Cohorts	Profit ($m)
More persistent	IV $<$ RV	461	-4.22
More persistent	IV $\geq$ RV	846	+2.93
More reversing	IV $<$ RV	409	-1.31
More reversing	IV $\geq$ RV	883	+6.09

A more reversing path has absolute entry-to-exit log return no greater than 20% of the sum of absolute daily log returns; other paths are classified as more persistent. Entry IV is the common annualized European, dividend-escrow volatility that reprices the combined call-and-put entry midpoint premium. Subsequent RV is the square root of 252 times mean squared daily log returns from entry close to

actual exit close, without subtracting the sample mean. Three cohorts exited early. The trading signal continues to use ATM-call IV; common straddle IV is an attribution measure only.

The IV–RV relation is not a profit identity: 220 cohorts lose money despite entry IV exceeding subsequent RV. Discrete hedging, jumps, changing gamma and execution charges matter. Classifications also depend on measurement conventions: calendar-time annualization changes 69 classifications, and a dividend adjustment changes six. Positive historical returns alone do not establish alpha after compensation for volatility risk.

D.3 Accounting Reconciliation

The 2,599 unique cohorts reconcile to the full-history funded account, including hedge trading costs, stock borrow, dividends and interest above the cash benchmark. One quote-availability convention was inferred; using recovered observed quotes changes aggregate hedge profit by $367.52. The retained figures use the reconciled saved-account attribution, and this reconstruction difference is not counted as a new profit source. Section A.5 states the limits of snapshot-based reproduction.